

AYUSH SHARMA

Software Engineer | Distributed Systems | Fintech, Risk & Lending Platforms

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PROFESSIONAL SUMMARY

Software engineer with **3+ years** designing and building **multi-tier distributed systems**, **Java / Spring Boot microservices** and **REST APIs**, for high-scale fintech platforms spanning fraud detection, credit underwriting, and BNPL lending, including real-time risk checks embedded directly in the **payment disbursal and transaction path**. Owns **8+ production services** processing **50K events/day** at **99.9% uptime**, providing **on-call production support** and RCAs while collaborating across engineering, QA, and release teams to roll out system changes at scale. Builds with **AI-assisted and agentic tooling** — designed an **MCP-based on-call agent** integrating Grafana, Slack, GitHub, and Jira that cut per-alert triage effort by **95%**. Works across the stack with **Java, Python, Spring Boot, SQL/NoSQL, Docker, Kubernetes**, and **CI/CD**.

TECHNICAL SKILLS

Languages:	Java, Python, Scala, JavaScript, SQL
Backend:	Distributed Systems, Microservices , Design Patterns, Spring Boot, REST APIs , Event-Driven Architecture, API Design
AI & GenAI:	Model Context Protocol (MCP) , Agentic AI workflows , Prompt engineering , AI-assisted coding (Claude)
Data & Streaming:	Apache Kafka , Akka Streams, Apache Airflow , Databricks
Databases:	PostgreSQL , DynamoDB , Redis , Elasticsearch , TimescaleDB (SQL & NoSQL)
Cloud & DevOps:	AWS (Lambda, EC2, DynamoDB), Docker , Kubernetes , Spinnaker, CI/CD
Observability:	Prometheus , Grafana , Kibana, Alerting, Incident Response

PROFESSIONAL EXPERIENCE

Software Development Engineer

June 2024 – Present

Jupiter Money — Risk Tech Team | Bangalore, India

- Designed and implemented **multi-tier distributed Java / Spring Boot microservices** and **REST APIs** with **CI/CD pipelines**, ingesting internal and third-party data at **50K events/day** for risk, fraud, and credit workflows.
- Owned **8+ production services**; maintained **99.9% SLA uptime** and provided **on-call production support** via rotation, alerting, and RCAs.
- Migrated **Airflow DAGs** from Athena to Databricks and cache from Elasticsearch to **Redis Cloud**; upgraded **Java services 17 to 21**, achieving **30% infrastructure cost reduction**.
- Fraud Detection Platform**
 - Integrated **4 third-party fraud signals** into real-time fraud-detection services — **Department of Telecom** (phone verification, blocking fraudulent mobile numbers), **BureauID** (device & identity fraud), **Sign3** (digital-footprint fraud), and **Finvu** (account aggregator).
 - Identified and blocked **30% of fraudulent users** at onboarding, cutting downstream credit losses while reducing false positives.
- Super Money, BNPL Lending Platform**
 - Built and owned core **Buy Now Pay Later (BNPL)** services end-to-end, expanding the lending business and driving **30% revenue growth**.
 - Engineered a **credit-line / limit engine** that computes per-user BNPL eligibility and credit limits from underwriting signals, powering real-time lending decisions.
 - Embedded **real-time risk and fraud checks** into the BNPL onboarding and transaction path, gating disbursals to curb defaults and fraudulent borrowing.
- CIBIL Credit-Profiling Integration**
 - Integrated **CIBIL** underwriting and credit-profiling signals, improving credit-decision accuracy — **cut false positives 20%** and **lifted true negatives 10%**.
 - Defined **API contracts** and coordinated cross-functional stakeholders to deliver production rollouts.
- MCP-Based Autonomous On-Call Agent**
 - Built an **agentic on-call assistant** integrating **Grafana, Slack, GitHub**, and **Jira** with Claude over the **Model Context Protocol (MCP)**, triggered automatically whenever an alert fires.
 - Automated **log and metric collection** across impacted, upstream, and downstream services, cutting per-alert manual triage by up to **95%**.
 - Generates a full **incident report per alert** — classifying **severity** and recommending **action items** — and **self-improves** by writing summaries of recurring alerts to memory, resolving them **autonomously** over time.

Software Development Engineering Intern

November 2023 – June 2024

Jupiter Money — Risk Tech Team | Bangalore, India

- Built **real-time ingestion pipelines** with **Kafka** and **Akka Streams** over **DynamoDB** and **PostgreSQL**, achieving **sub-100ms latency**.
- Developed **5–6 Airflow DAGs**, **3–4 AWS Lambdas**, and **Spring Boot APIs**; cut deployment automation time from **30 minutes to 5 minutes**.
- Implemented production monitoring (**Prometheus, Grafana, Kibana**) for critical pipelines and supported launches, incident triage, and **RCAs**.

KEY PROJECTS

Metrics Publisher Service [GitHub]

Java 17, Spring Boot, Apache Kafka, TimescaleDB, Redis, Docker, Gradle

- Engineered a distributed, high-throughput **time-series metrics collection and query system** with a **4-stage event-driven pipeline**.
- Implemented **batch processing** (500 metrics/batch), **Redis-based cardinality control** (1K series/metric), and **T-Digest quantile estimation**.
- Exposed **REST API** for P95, SUM, AVG, RATE aggregations with time-bucketed queries, 90-day retention, and compression.

Movie Ticket Booking System [GitHub]

Java 21, Spring Boot 3, PostgreSQL, Flyway, Spring Security, Testcontainers, Gradle

- Built a **seat-level booking backend** as a package-by-feature **modular monolith** — time-bound holds, tiered/weekend pricing, discount codes, simulated payment, and configurable refunds.
- Serialized concurrent seat booking with a **lock-free guarded conditional UPDATE** in PostgreSQL; a **100-thread race test** yields exactly one winner with **zero double-allocation**.
- Ran payment I/O outside DB transactions with **idempotency keys** and non-blocking **transactional-outbox** notifications; **85 tests** green against real Postgres (Testcontainers).

EDUCATION

B.Tech, Computer Science and Engineering

2020 – 2024

Indraprastha Institute of Information Technology, Delhi (IIITD) | CGPA: 8.0 / 10

CERTIFICATIONS, RECOGNITION & ADDITIONAL

- Best Performer of the Quarter** — Jupiter Money (Awarded for consistent delivery and production ownership)
- ARTPARK Fellowship** — IISc Bangalore (Awarded for research contributions)
- Amazon ML Summer School** — Amazon (Competitively selected), 2022
- Teaching Assistant** — Basic Electronics (300+ undergraduates) at IIITD